

BUSINESS INTELLIGENCE ANALYSIS OF SUPERSTORE SALES

Using SQL and Microsoft Power BI Independent Data Analytics Project Report

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Tools and Technologies Used

- Microsoft SQL Server
- Microsoft Power BI
 - Microsoft Excel
- Visual Studio Code

Project Focus

Business Intelligence • SQL Data Analysis • Data Visualization • Business Analytics

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CHAPTER 1 : Introduction

1.1 Background of the Study

In today's competitive business environment, organizations generate large amounts of sales data through their daily operations. However, collecting data alone is not enough to support effective decision-making. Businesses need to analyze this information to identify sales trends, customer purchasing behavior, product performance, and regional differences. Data analysis helps managers understand business performance and make informed decisions that improve profitability and operational efficiency.

Structured Query Language (SQL) is one of the most widely used tools for managing and analyzing business databases. It allows analysts to retrieve, filter, summarize, and organize large datasets efficiently. By using SQL, businesses can convert raw transaction records into meaningful information that supports strategic planning and decision-making.

This project analyzes the Sample Superstore dataset using SQL to examine different aspects of sales performance, including customer behavior, product categories, profitability, and regional performance. Through a series of business-oriented SQL queries, the project generates insights that can help management identify strengths, recognize potential areas for improvement, and support future business decisions.

The project also demonstrates practical SQL skills that are widely used in business analytics and data analysis. It highlights how SQL can be applied to solve real-world business problems while transforming raw business data into valuable insights.

1.2 Problem Statement

Businesses generate a large amount of sales data every day, but making effective use of this data remains a challenge. Many organizations collect information about customers, products, sales, and profits, yet they often struggle to identify useful patterns that can support better business decisions. Without proper analysis, it

becomes difficult to understand which products perform well, which regions generate the highest profit, and which customers contribute most to overall sales.

SQL provides an effective way to organize and analyze business data. However, simply writing SQL queries is not enough. The results must also be interpreted to produce meaningful business insights that can help managers improve performance and make informed decisions.

This project addresses this problem by using SQL to analyze the Sample Superstore dataset. Through a series of business questions, the project examines sales performance, profitability, customer behavior, and regional trends. The findings aim to provide practical insights that demonstrate how SQL can support data-driven decision-making in a business environment.

1.3 Objectives of the Study

The main objective of this project is to analyze the Sample Superstore dataset using SQL and generate meaningful business insights that can support better decision-making.

The specific objectives of the study are:

To analyze the overall sales performance of the company.

To identify the most profitable products, categories, and sub-categories.

To evaluate sales and profit across different states and regions.

To identify the top customers based on total sales.

To examine the impact of discounts on sales and profitability.

To analyze customer purchasing patterns using SQL queries.

To develop practical recommendations based on the analysis results.

1.4 Research Questions

This project aims to answer the following research questions:

1. Which product categories generate the highest sales and profit?
2. Who are the top customers based on total sales?
3. Which states and regions contribute the most to overall sales and profitability?
4. How do discounts affect sales and profit?
5. Which products perform best in terms of sales and profitability?
6. What business insights can be generated through SQL-based data analysis?
7. How can the findings help improve business decision-making?

1.5 Scope of the Study

This project focuses on analyzing the Sample Superstore dataset using SQL. The analysis is limited to the information available in the dataset, including customer details, product categories, sales, profit, discounts, shipping modes, and regional performance. The project examines business trends by answering a series of SQL-based business questions and interpreting the results.

The findings are intended to provide insights into sales performance and profitability. Since the analysis is based on a sample dataset, the recommendations are for learning and analytical purposes rather than representing the performance of a real company.

1.6 Significance of the Study

This project demonstrates how SQL can be used to analyze business data and support data-driven decision-making. It highlights the importance of organizing and interpreting sales data to identify business trends and improve overall performance. From an academic perspective, the project strengthens practical knowledge of SQL and business analytics by applying database concepts to real-world business scenarios. It also develops analytical and problem-solving skills through the interpretation of sales and profitability data.

From a professional perspective, the project reflects the ability to work with business datasets, write SQL queries, and communicate findings effectively. These skills are valuable in fields such as data analytics, business intelligence, finance, and business analysis.

CHAPTER 2: Dataset Description

2.1 Dataset Overview

The analysis in this project is based on the Sample Superstore dataset, which is commonly used for learning business analytics and SQL. The dataset contains transaction-level sales records of a retail company and provides detailed information about customers, products, sales, profits, discounts, shipping methods, and geographical locations.

Each record in the dataset represents a single customer order. The available information allows different aspects of business performance to be analyzed, including customer purchasing behavior, regional sales, product profitability, and overall business performance.

The dataset was imported into an SQLite database and analyzed using SQL queries. The results were then interpreted to generate business insights and recommendations.

2.2 Dataset Variables

Variable	Description
Order ID	Unique identification number for each order
Order Date	Date on which the order was placed
Ship Date	Date on which the order was shipped
Ship Mode	Shipping method used
Customer ID	Unique customer identifier
Customer Name	Name of the customer
Segment	Customer segment
Country	Customer's country
City	Customer's city
State	Customer's state
Region	Business region
Product ID	Unique product identifier
Category	Product category
Sub-Category	Product sub-category
Product Name	Name of the product
Sales	Revenue generated from the sale
Quantity	Number of units sold
Discount	Discount applied to the order
Profit	Profit earned from the transaction

2.3 Data Preparation:

Before performing the analysis, the dataset was imported into SQLite and reviewed to ensure that all records were available for analysis. The data structure was examined, and SQL queries were executed on the original dataset without modifying the transaction records. The analysis focused on retrieving, grouping, filtering, sorting, and summarizing data to answer business-related questions.

The SQL analysis included the use of aggregate functions, filtering conditions, grouping operations, ordering techniques, and ranking methods to generate meaningful business insights.

CHAPTER 3: Methodology

3.1 Research Method

This project follows a quantitative analytical approach because it is based on numerical business data. The analysis was carried out using SQL to examine sales transactions and answer business-related questions. Rather than collecting primary data through surveys or interviews, the study uses an existing business dataset to identify patterns, trends, and relationships.

The project focuses on analyzing sales performance, profitability, customer behavior, and regional performance through structured SQL queries. The results are interpreted to provide practical business insights and recommendations.

3.2 Software and Tools Used

Tool	Purpose
SQLite	Database Management System
DB Browser for SQLite	Running SQL Queries
Visual Studio Code	Writing SQL Script
Microsoft Word	Report Writing

3.3 SQL Techniques Used:

Different SQL techniques were applied throughout the project to answer business questions and analyze the dataset. These techniques helped retrieve, summarize, and organize the data into meaningful information.

The main SQL techniques used include:

- SELECT Statement
- WHERE Clause
- ORDER BY
- GROUP BY
- Aggregate Functions (SUM, AVG, COUNT, MAX, MIN)
- LIMIT
- ROUND Function
- Aliases (AS)
- These techniques were used to perform business analysis and generate insights from the sales data.

3.4 Analysis Procedure

The project was completed following a systematic process to ensure that the analysis was organized and easy to understand.

The main steps included:

1. Importing the Sample Superstore dataset into SQLite.
2. Creating and managing the database using DB Browser for SQLite.
3. Writing SQL queries to answer business-related questions.
4. Executing each query and reviewing the output.
5. Interpreting the results through observations.
6. Providing business recommendations based on the findings.
7. Presenting the overall analysis using a dashboard and final report.

CHAPTER 4: SQL Analysis and Business Findings

4.1 Introduction

This chapter presents the results of the SQL analysis conducted on the Sample Superstore dataset. SQL queries were used to retrieve, summarize, and analyze business information related to sales performance, profitability, customer behaviour,

product categories, regional performance, and shipping methods. Each business question was answered using SQL statements, and the results were interpreted to identify meaningful insights that can support managerial decision-making.

The analysis focuses not only on obtaining numerical results but also on understanding their business implications. Each query is followed by an observation, interpretation, and recommendation to demonstrate how SQL can be applied to solve real business problems through data analysis.

4.2 Sales Performance Analysis

The SQL analysis indicates that the company generated strong overall sales during the study period. The total sales exceeded \$2.29 million, reflecting a high level of business activity across multiple product categories and customer segments. Analysis of category-wise sales revealed that Technology products contributed the highest share of revenue, followed by Furniture and Office Supplies.

The analysis also highlighted noticeable differences in sales performance across product sub-categories. While several products generated consistently high revenue, others contributed relatively little to total sales. These differences suggest that customer preferences vary significantly across product types and that product demand is not evenly distributed throughout the inventory.

4.3 Profitability Analysis

Although the company generated substantial sales revenue, the SQL analysis showed that overall profitability was considerably lower than total sales. This indicates that a significant portion of revenue is absorbed by operational costs, product costs, and discounts offered to customers. Therefore, higher sales do not necessarily translate into higher profit.

The profitability analysis further revealed that profit varied across product categories and regions. Some products generated strong revenue but relatively lower profit margins, while others contributed positively to profitability despite lower sales

volumes. This demonstrates the importance of evaluating both sales and profit simultaneously when assessing business performance.

Understanding profitability helps managers identify products and business segments that contribute most effectively to financial growth and supports better pricing and inventory decisions.

4.4 Customer and Order Analysis

The SQL analysis identified a large number of customer orders within the dataset, indicating consistent customer purchasing activity throughout the study period. The business serves hundreds of unique customers across different market segments, reflecting a broad customer base and diversified demand.

Analysis of customer behaviour showed that purchasing patterns differ among customers, with some contributing significantly more revenue than others. Such information is valuable for identifying loyal customers and designing customer retention strategies. Monitoring customer activity also enables businesses to evaluate sales performance more effectively and improve long-term customer relationships.

4.5 Product Performance Analysis

Product-level analysis provided valuable insights into the performance of individual categories and sub-categories. Technology products emerged as the strongest contributors to total sales, while several sub-categories consistently outperformed others in terms of revenue generation.

The analysis also identified products with comparatively lower sales performance. These products may require additional promotional efforts, pricing adjustments, or inventory evaluation to improve their contribution to overall business performance.

Evaluating product performance enables businesses to allocate resources more efficiently and prioritize investments in high-performing product lines.

4.6 Regional Analysis

The SQL analysis revealed differences in sales performance across geographical regions. Certain regions generated significantly higher sales and profit than others, indicating variations in customer demand, market size, and purchasing behaviour.

Regional analysis allows management to identify strong-performing markets while recognizing areas where additional marketing efforts or operational improvements may be required. Understanding geographical performance also supports better resource allocation and expansion planning.

4.7 Shipping Analysis

Shipping analysis examined customer preferences for different shipping methods used by the company. The results indicate that some shipping modes are utilized more frequently than others, reflecting customer priorities regarding delivery speed, convenience, and cost.

An understanding of shipping preferences enables businesses to optimize logistics operations, improve delivery efficiency, and maintain customer satisfaction while controlling transportation costs.

4.8 Chapter Summary

The SQL analysis successfully transformed raw transactional data into meaningful business information. The findings demonstrated strong overall sales performance, positive profitability, active customer participation, and significant variations in product and regional performance. These insights provide valuable evidence for business decision-making and establish the analytical foundation for the interactive Power BI dashboard presented in the next chapter.

CHAPTER 5: Dashboard Analysis

5.1 Dashboard Overview

To improve the presentation and interpretation of analytical results, an interactive dashboard was developed using Microsoft Power BI. The dashboard was designed to transform the SQL analysis into visual representations that allow users to understand

business performance quickly and effectively. Rather than interpreting raw numerical values, decision-makers can evaluate important business indicators through charts, graphs, and Key Performance Indicators (KPIs).

The dashboard combines multiple visualizations to provide a comprehensive overview of sales performance, profitability, customer activity, regional performance, product categories, and shipping methods. Every visualization was carefully selected to answer a specific business objective and to simplify complex business information into an easily understandable format.

The interactive nature of Power BI enables users to monitor business performance from different perspectives while supporting faster and more informed managerial decisions. The dashboard serves as a practical business intelligence tool that demonstrates how data visualization enhances organizational decision-making.

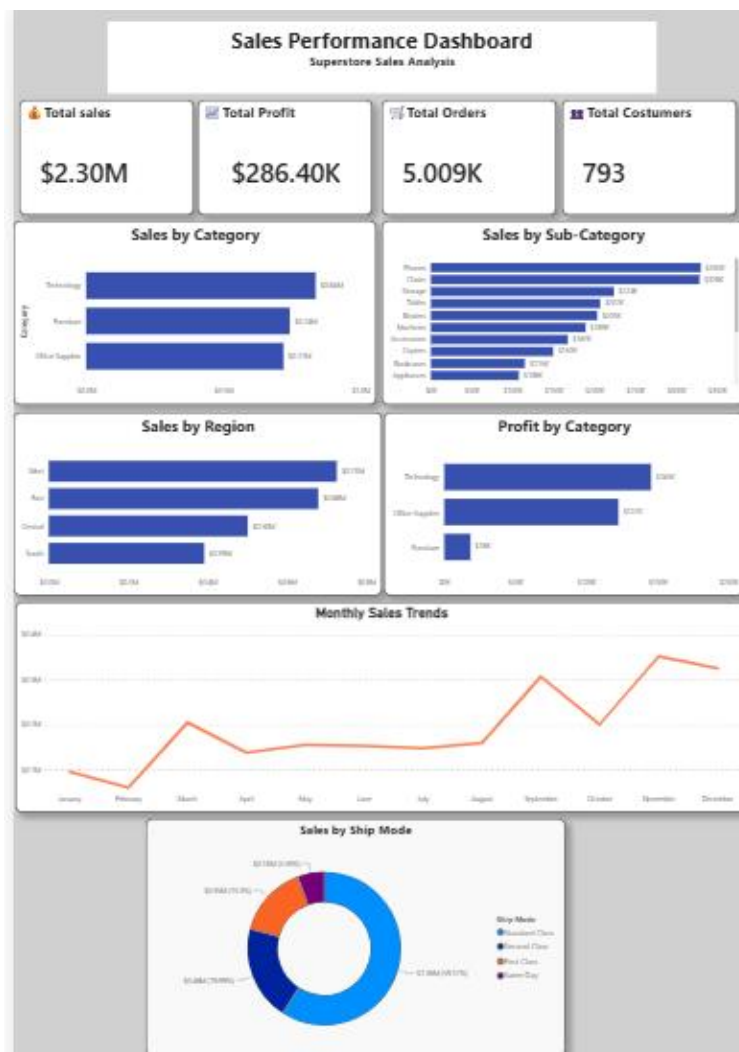


Figure 5.1: Sales Performance Dashboard Developed Using Microsoft Power BI

5.2 Dashboard Components

The dashboard consists of four Key Performance Indicators (KPIs) and six analytical visualizations. Together, these components provide a complete summary of the company's business performance and allow users to identify important trends, patterns, and relationships within the sales data.

The dashboard includes the following components:

- ◆ Total Sales
- ◆ Total Profit
- ◆ Total Orders
- ◆ Total Customers
- ◆ Sales by Category
- ◆ Sales by Sub-Category
- ◆ Sales by Region
- ◆ Profit by Category
- ◆ Monthly Sales Trend
- ◆ Sales by Ship Mode

Each component contributes to understanding a different aspect of the business and collectively supports effective managerial decision-making.

5.3 Key Performance Indicators (KPIs)

- **Total Sales**

The Total Sales KPI represents the total revenue generated from all transactions included in the dataset. The dashboard reports total sales of approximately \$2.30 million, indicating strong business activity during the study period. This KPI provides an overall measure of revenue generation and serves as the primary indicator of organizational performance.

- **Total Profit**

The Total Profit KPI measures the company's earnings after considering the profit generated from individual transactions. The dashboard reports an overall profit of approximately \$286 thousand, demonstrating that the company maintained positive financial performance throughout the study period. Comparing total sales with total profit also provides valuable insight into the company's overall profitability.

- **Total Orders**

The Total Orders KPI indicates the total number of customer orders processed during the study period. A large number of completed orders reflects consistent customer purchasing activity and demonstrates the company's ability to maintain stable business operations across different product categories.

- **Total Customers**

The Total Customers KPI represents the number of unique customers served by the company. This indicator highlights the size of the customer base and provides useful information for evaluating customer reach and market coverage. A diverse customer base contributes to business stability and reduces dependence on a limited number of buyers.

5.4 Dashboard Visualizations

5.4.1 Sales by Category

The Sales by Category chart compares total sales across the three major product categories. The visualization shows that the Technology category generated the highest sales, followed by Furniture and Office Supplies. This indicates that technology products are the primary source of revenue for the company and should remain an important strategic focus.

5.4.2 Sales by Sub-Category

The Sales by Sub-Category visualization provides a more detailed breakdown of product performance. It highlights the individual product groups that contribute most significantly to total sales while identifying lower-performing sub-categories. This information helps management prioritize inventory planning, product promotion, and sales strategies.

5.4.3 Sales by Region

The Sales by Region chart compares sales performance across different geographical regions. The visualization demonstrates noticeable differences in regional sales, indicating variations in customer demand and market potential. Understanding regional performance enables businesses to allocate resources more effectively and develop region-specific marketing strategies.

5.4.4 Profit by Category

The Profit by Category chart illustrates how profit is distributed among the major product categories. While high sales contribute to overall revenue, profitability differs across categories due to pricing strategies, discounts, and product costs. This visualization helps management evaluate financial performance beyond sales volume alone.

5.4.5 Monthly Sales Trend

The Monthly Sales Trend line chart illustrates changes in sales over time. The visualization allows management to identify seasonal patterns, periods of high demand, and fluctuations in business performance. Monitoring monthly trends supports improved forecasting, production planning, and inventory management.

5.4.6 Sales by Ship Mode

The Sales by Ship Mode visualization presents the distribution of sales across different shipping methods. The results indicate customer preferences regarding delivery services and provide useful information for evaluating logistics performance.

Understanding shipping patterns can help businesses improve delivery efficiency while maintaining customer satisfaction.

5.5 Chapter Summary

The Power BI dashboard successfully transformed the SQL analysis into an interactive and visually appealing business intelligence tool. The dashboard provides a comprehensive overview of sales, profitability, customer activity, regional performance, product performance, and shipping methods through a combination of KPIs and analytical visualizations. These visual insights enhance business understanding and support more effective strategic decision-making. The dashboard demonstrates the practical application of Power BI in communicating analytical findings and improving organizational performance monitoring.

CHAPTER 6: Overall Findings

6.1 Overall Findings

The analysis conducted using SQL and Microsoft Power BI provided valuable insights into the overall business performance of the Sample Superstore dataset. The findings demonstrate that data analytics can effectively transform raw business data into meaningful information that supports strategic decision-making. By combining SQL for data extraction and Power BI for visualization, the project successfully identified important trends related to sales, profitability, customer behavior, regional performance, product categories, and shipping methods.

The company generated total sales of approximately \$2.30 million, indicating a strong sales performance during the study period. However, the analysis also revealed that high sales do not necessarily guarantee high profitability. Although the company earned a positive total profit of approximately \$286 thousand, the difference between total sales and profit suggests that operational costs, discounts, and pricing strategies significantly influence financial performance.

Product category analysis showed that the Technology category generated the highest sales and served as the primary contributor to overall revenue. In contrast, other categories contributed comparatively lower sales, highlighting differences in customer demand across product groups. Analysis at the sub-category level further identified products with strong market performance while also revealing products requiring additional marketing attention.

Regional analysis demonstrated that sales performance varied across geographical locations. Certain regions consistently generated higher sales and profits than others, indicating differences in market potential and customer purchasing behavior. These findings emphasize the importance of region-specific business strategies and efficient resource allocation.

Customer analysis revealed a diverse customer base with a high number of completed orders throughout the study period. Consistent customer activity reflects stable business operations and demonstrates the company's ability to maintain long-term market engagement. Understanding customer purchasing patterns enables businesses to strengthen customer relationships and improve future marketing strategies.

The monthly sales trend illustrated fluctuations in sales throughout the year, suggesting the presence of seasonal demand patterns. Such information is valuable for forecasting future sales, planning inventory requirements, and preparing promotional campaigns during periods of lower business activity.

Finally, the Power BI dashboard successfully converted the analytical findings into an interactive business intelligence tool. The dashboard provides management with a clear overview of organizational performance through KPIs and visualizations, enabling faster interpretation of complex business information and supporting evidence-based decision-making.

Overall, the findings confirm that combining SQL and Power BI provides an effective framework for business data analysis. The project demonstrates how analytical tools can help organizations monitor performance, identify opportunities for improvement, and support strategic business planning.

6.2 Key Findings

The major findings of this project are summarized below:

- ✧ The company generated total sales of approximately \$2.30 million during the study period.
- ✧ Overall profit remained positive at approximately \$286 thousand, although profit growth was lower than total sales.
- ✧ The Technology category generated the highest sales among all product categories.
- ✧ Sales performance varied significantly across different product sub-categories.
- ✧ Regional analysis revealed noticeable differences in market performance across geographical areas.
- ✧ The company maintained a large customer base with consistent purchasing activity.
- ✧ Monthly sales trends indicated fluctuations in customer demand throughout the year.
- ✧ Different shipping methods contributed differently to overall sales performance.
- ✧ SQL proved effective for extracting and summarizing business information.
- ✧ Power BI enhanced the presentation of analytical findings through interactive visualizations and KPIs.

CHAPTER 7: Business Recommendations

7.1 Business Recommendations

Based on the SQL analysis and Power BI dashboard findings, several recommendations can be made to improve the company's overall business performance. These recommendations focus on increasing sales, improving profitability, strengthening customer relationships, and supporting long-term business growth.

Recommendation 1: Strengthen High-Performing Product Categories

The Technology category generated the highest sales during the analysis period, making it the company's strongest revenue source. Management should continue investing in this category by maintaining sufficient inventory, introducing new products, and implementing targeted promotional campaigns. Focusing on high-performing categories can further increase overall revenue.

Recommendation 2: Improve the Performance of Low-Selling Products

Some product categories and sub-categories generated comparatively lower sales than others. These products should be evaluated to identify the reasons behind their weaker performance. Management may consider revising pricing strategies, offering promotional discounts, improving product visibility, or discontinuing consistently underperforming items if necessary.

Recommendation 3: Focus on Profitability Alongside Sales

Although total sales were relatively high, overall profitability was considerably lower. This indicates that increasing sales alone is not sufficient to maximize business performance. Management should regularly monitor profit margins, control unnecessary costs, and reduce excessive discounting to improve overall financial performance.

Recommendation 4: Develop Region-Specific Business Strategies

The analysis revealed noticeable differences in sales performance across geographical regions. Management should allocate marketing resources according to regional demand and develop customized sales strategies for lower-performing regions. Expanding successful business practices from high-performing regions may improve sales in weaker markets.

Recommendation 5: Strengthen Customer Retention Programs

The company serves a large customer base with consistent purchasing activity. Developing customer loyalty programs, personalized marketing campaigns, and after-

sales services can improve customer satisfaction and encourage repeat purchases. Retaining existing customers is often more cost-effective than acquiring new ones.

Recommendation 6: Improve Inventory Management

The identification of high-performing products provides valuable information for inventory planning. Frequently purchased products should remain adequately stocked to avoid shortages, while inventory levels for slow-moving products should be carefully monitored to reduce storage costs and unnecessary stock accumulation.

Recommendation 7: Utilize Seasonal Sales Trends

Monthly sales trends indicate fluctuations in customer demand throughout the year. Management should use these seasonal patterns to improve demand forecasting, inventory planning, and promotional activities. Special marketing campaigns during slower months may help stabilize overall sales performance.

Recommendation 8: Optimize Shipping Operations

Different shipping methods contribute differently to business performance and customer satisfaction. The company should regularly evaluate shipping efficiency, delivery times, and transportation costs to improve logistics performance while maintaining high customer satisfaction levels.

Recommendation 9: Implement Continuous Business Monitoring

Rather than conducting business analysis only once, the company should continuously monitor sales performance through interactive dashboards and periodic SQL reports. Regular performance monitoring enables management to identify emerging trends, detect operational problems early, and make timely business decisions.

Recommendation 10: Promote Data-Driven Decision Making

Organizations should encourage the use of analytical tools such as SQL and Microsoft Power BI in routine business operations. Data-driven decision-making reduces uncertainty, improves planning accuracy, and enables managers to make objective decisions based on reliable business evidence rather than assumptions.

7.2 Chapter Summary

This chapter presented practical business recommendations based on the findings obtained from the SQL analysis and Power BI dashboard. The recommendations focus on improving sales performance, increasing profitability, strengthening customer relationships, optimizing inventory and logistics operations, and encouraging data-driven decision-making. Implementing these recommendations can support sustainable business growth while improving overall organizational performance.

CHAPTER 8: Conclusion

8.1 Conclusion

This project successfully demonstrated the practical application of SQL and Microsoft Power BI in analyzing business data and transforming raw information into meaningful business insights. The primary objective of the study was to evaluate the sales performance of the Sample Superstore dataset and develop an interactive dashboard that supports data-driven decision-making. Through systematic data analysis, the project highlighted the importance of combining database management with data visualization to solve real-world business problems.

SQL was used to organize, retrieve, and summarize transactional data through various analytical techniques, including filtering, grouping, sorting, aggregate functions, and conditional statements. These techniques made it possible to evaluate business performance from multiple perspectives, including sales, profitability, customer activity, product performance, regional distribution, and shipping methods. The SQL analysis provided a structured understanding of the dataset and established the analytical foundation for further visualization.

Microsoft Power BI was then utilized to transform the analytical results into an interactive dashboard containing key performance indicators and business visualizations. The dashboard simplified complex business information into an easy-to-understand format, enabling users to evaluate organizational performance more efficiently. The visualizations provided valuable insights into sales trends, product

categories, customer behavior, and regional performance while supporting faster interpretation of business information.

The findings indicate that the company achieved strong overall sales performance; however, profitability remained significantly lower than total revenue, emphasizing the importance of managing operational costs alongside increasing sales. Product category analysis identified Technology as the strongest-performing category, while regional and customer analyses highlighted opportunities for targeted business strategies. The dashboard further demonstrated how visual analytics can improve communication of business information and support strategic planning.

Overall, this project illustrates the growing importance of business intelligence and data analytics in modern organizations. The integration of SQL and Power BI provides an effective approach for analyzing business performance, identifying operational opportunities, and supporting evidence-based decision-making. The project also enhanced practical skills in database management, data analysis, dashboard development, and business reporting, making it a valuable learning experience and an important addition to a professional analytics portfolio.

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